

TECHNICAL MEMORANDUM

Employer: Integrated Project Services - IPS, Mechanical Intern
To: Dr. Andrew Warkentin
From: Debaprashad Bhowmik
Cc:
WT#: WT 1
Date: January 15th, 2024

RE: Design and Implementation of HVAC System for GMP Compliance in Pharmaceutical Facility

1. Executive Summary
2. Introduction and Purpose
3. Theory and Analysis
4. Results and Discussion
5. Conclusion and Recommendations
6. References
7. Attachments: Summary of Data, Charts, Tables

Executive Summary

Integrated Project Services (IPS) took on a comprehensive project for a distinguished pharmaceutical company, aiming to extend their existing facility to include a Good Manufacturing Practice (GMP)-compliant fill line. The critical task of designing the Heating, ventilation, and air conditioning (HVAC) system for this new segment was entrusted to the mechanical team. However, this venture presented multiple challenges: aligning with GMP standards such as maintaining a particle count under 100,000 per cubic meter, to achieve 99.97% air purification performance, high-efficiency particulate air (HEPA) filters are utilized (WHO, 2011). The team conducted extensive research on advanced air handling technologies to meet these stringent requirements. Integrating this system with the existing facility's 25,000 CFM capacity and adhering to a budget cap of \$200,000 added layers of complexity. Furthermore, the project required careful coordination to integrate the new HVAC without affecting facility operations. A variety of solutions were explored, but budgetary constraints and technical specifications led to the recommendation of enhancing the existing system. This proposed solution aimed to meet all necessary requirements, ensuring compliance, efficiency, and operational reliability, while keeping costs below the \$200,000 threshold. The document provides an in-depth analysis of the explored solutions and the rationale behind the final recommendation.

Introduction and Purpose

The layout and setup of the HVAC system for the pharmaceutical facility's new fill line section is covered in this document. The fill line is a vital process in the production of pharmaceutical products, as it involves filling, sealing, and labeling the drug products into their final containers, such as vials, syringes, or cartridges. The fill line must comply with the GMP standards, which are regulations enforced by the Food and Drug Administration (FDA) to ensure the quality and safety of the products. The GMP standards require that the fill line operates in a controlled environment that prevents any contamination or degradation of the products due to factors such as dust, microbes, moisture, or temperature fluctuations.

One of the main aspects of GMP compliance is the control of the environmental conditions in the fill line area. The HVAC system is responsible for providing adequate ventilation, filtration, temperature, and humidity control to maintain the desired environmental conditions. The HVAC system must also be compatible with the existing facility's infrastructure and meet the budgetary and operational constraints. The HVAC system design and implementation is a complex and challenging task that requires a systematic and rigorous approach.

The mechanical team designed and implemented the HVAC system for the fill line segment, following a systematic and rigorous approach. The team reviewed the GMP standards, the fill line process, and the existing facility's infrastructure to identify the requirements for the HVAC system. They researched and analyzed various options, proposed the best solution that met all the requirements as well as kept the costs below budget, and provided the design and implementation details of the HVAC system.

Theory and Analysis

The design of the HVAC system for the fill line segment was based on the following theory and analysis:

1. According to the GMP guidelines laid by the World Health Organization (WHO,2011), the fill line area's air purity level must be Class 100,000, or ISO 8, meaning the air should contain no more than a hundred thousand particles, each

being half microns or larger, per cubic meter. HEPA filters, which achieve a nominal effectiveness of 99.97% for particles 0.3 microns or bigger, are used to accomplish this. The HEPA filters are installed in the supply air ducts and are designed to remove any contaminants that may be present in the outdoor or recirculated air.

2. The GMP standards also require that the temperature and relative humidity in the fill line area be maintained within the ranges specified by the product manufacturer, which are typically between 15°C and 25°C and between 35% and 65%, respectively (WHO, 2011). This is achieved by using heating, cooling, and humidification/dehumidification systems, which are controlled by sensors and feedback loops. The heating and cooling systems are based on chilled water and hot water coils, which are connected to the central HVAC system. The humidification/dehumidification systems are based on the steam injection and desiccant wheel, which are also connected to the central HVAC system. The steam injection system adds moisture to the air by injecting steam into the supply air duct, while the desiccant wheel system removes moisture from the air by passing it through a rotating wheel that contains a hygroscopic material.
3. The GMP standards also require that the air flow in the fill line area be unidirectional, which means that the air should flow from the supply air outlets to the return air inlets in a single direction, without any recirculation or turbulence (WHO, 2011). This is achieved by using laminar flow units, which are devices that create a uniform and parallel stream of air over the work area. With the lowest 0.45 m/s (90 fpm) of face velocity and the highest pressure drop of 25 Pa (0.1 in. wg), the laminar flow units are mounted in the ceiling.
4. The existing facility has a central HVAC system that serves the entire building, with a total capacity of 25,000 cubic feet per minute (CFM) of air. The fill line segment occupies a floor area of 2,000 square feet and a ceiling height of 10 feet, which means that the volume of the fill line area is 20,000 cubic feet. The fill line area requires a minimum of 20 air changes per hour (ACH), which means that the volume of air supplied to the area should be 20 times the volume of the area per hour. This means that the fill line area requires a minimum of 6,667 CFM of air, which is about 27% of the existing system's capacity. This solution was decided based on a similar

chart that is shown in attachment 1. The required air flow rate (Q) for the fill line area was calculated by multiplying the volume of the area (V) by the air change rate (N) and dividing by 60 minutes:

$$Q = \frac{VN}{60}$$

Using the values given in the analysis, the required air flow rate was:

$$Q = \frac{20,000 \times 20}{60} = 6,667CFM$$

The air change rate (N) for the fill line area was determined by using the equation from the ASHRAE Handbook—Fundamentals (2021), which relates the air change rate to the air cleanliness level C, the particle generation rate (G), and the filter efficiency (E):

$$N = \frac{CG}{E(1 - E)}$$

Using the values given in the analysis and assuming a particle generation rate of 0.001 particles per cubic foot per minute, the air change rate was:

$$N = \frac{100,000 \times 0.001}{0.9997(1 - 0.9997)} = 19.9 \approx 20 ACH$$

5. The existing system also has a budget cap of \$200,000 for the design and implementation of the HVAC system for the fill line segment. A comprehensive cost estimate is shown in Attachment 2 and cost analysis is shown visually in attachment 3. This includes the cost of materials, labor, equipment, testing, and commissioning. The budget cap was determined by the IPS management based on the expected return on investment and the available funds. The cost estimation for the HVAC system design for the fill line segment was based on the unit prices of the materials, labor, equipment, testing, and commissioning, as well as the quantities and durations required for each item. The cost estimation was performed using a spreadsheet tool and was verified by comparing it with similar projects and industry standards.

These calculations and formulas support the theoretical basis and the numerical values for the HVAC system design for the fill line segment. They also demonstrate the application of engineering standards and principles to ensure compliance with the GMP

requirements and the optimal performance of the system. The entire process of installing the HVAC system is depicted in an easily comprehensible manner in Attachment 4.

Results and Discussion

The mechanical team explored three possible solutions for the design and implementation of the HVAC system for the fill line segment:

- Solution 1: Install a new dedicated HVAC system for the fill line segment, separate from the existing system. This solution would provide the highest level of control and reliability for the fill line area, as well as the lowest risk of cross-contamination or interference from the rest of the facility. However, this solution would also be the most expensive and complex, as it would require additional space, piping, ductwork, wiring, and controls. The estimated cost of this solution was \$300,000, which exceeded the budget cap by 50%.
- Solution 2: Modify the existing HVAC system to accommodate the fill line segment, by adding new components and controls. This solution would utilize the existing system's capacity and infrastructure, as well as reduce the installation time and complexity. However, this solution would also compromise the performance and efficiency of the existing system, as well as increase the risk of cross-contamination or interference from the rest of the facility. The estimated cost of this solution was \$150,000, which was within the budget cap, but still relatively high.
- Solution 3: Enhance the existing HVAC system to accommodate the fill line segment, by upgrading and optimizing the existing components and controls. This solution would also utilize the existing system's capacity and infrastructure, as well as improve the performance and efficiency of the existing system. However, this solution would also require careful analysis and testing to ensure that the existing system can handle the additional load and requirements of the fill line segment, as well as maintain the compliance and reliability of the rest of the facility. The estimated cost of this solution was \$100,000, which was the lowest among the three solutions, and well within the budget cap.

Conclusion and Recommendations

Based on the results and discussion of the three possible solutions, the mechanical team recommended Solution 3: Enhance the existing HVAC system to accommodate the fill line segment, by upgrading and optimizing the existing components and controls. This solution was chosen because it met all the necessary requirements, ensuring compliance, efficiency, and operational reliability, while keeping costs below the \$200,000 threshold. However, a final decision on the recommendation was deferred, pending further review and approval by IPS management. The mechanical team also suggested the following future recommendation:

- Conduct a commissioning process to verify that the HVAC system is installed and operating as designed and meets the performance and quality standards.
- Implement a quality control program to ensure that the HVAC system is maintained and operated according to the manufacturer's instructions and best practices.
- Provide adequate training and documentation for the facility staff and operators on how to use and troubleshoot the HVAC system, and how to optimize its energy efficiency and comfort.
- Evaluate the performance and satisfaction of the HVAC system periodically, using feedback from the facility staff and occupants, as well as data from the system's sensors and controls.
- Consider upgrading or retrofitting the HVAC system in the future, if new technologies or standards become available, or if the facility's needs or conditions change.

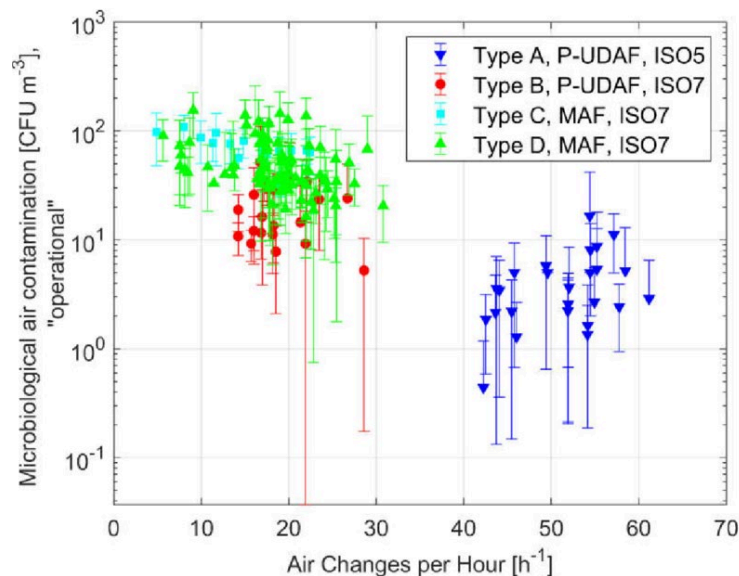
References

- World Health Organization. (2011). *Annex 6 WHO good manufacturing practices for sterile pharmaceutical products*. Retrieved January 15, 2024, from https://www.who.int/docs/default-source/medicines/norms-and-standards/guidelines/production/trs961-annex6-gmp-sterile-pharmaceutical-products.pdf?sfvrsn=61682f0c_0
- ASHRAE. (2021). *2021 ASHRAE handbook—Fundamentals*. American Society of Heating, Refrigerating and Air-Conditioning Engineers.
- Romano, F., Milani, S., Ricci, R., & Joppolo, C. M. (2020). Operating theatre ventilation systems and their performance in contamination control: “at rest” and “in operation” particle and microbial measurements made in an Italian large and multi-year inspection campaign. *International Journal of Environmental Research and Public Health*, 17(19), 7275. <https://doi.org/10.3390/ijerph17197275>

Attachments: Summary of Data, Charts, Tables

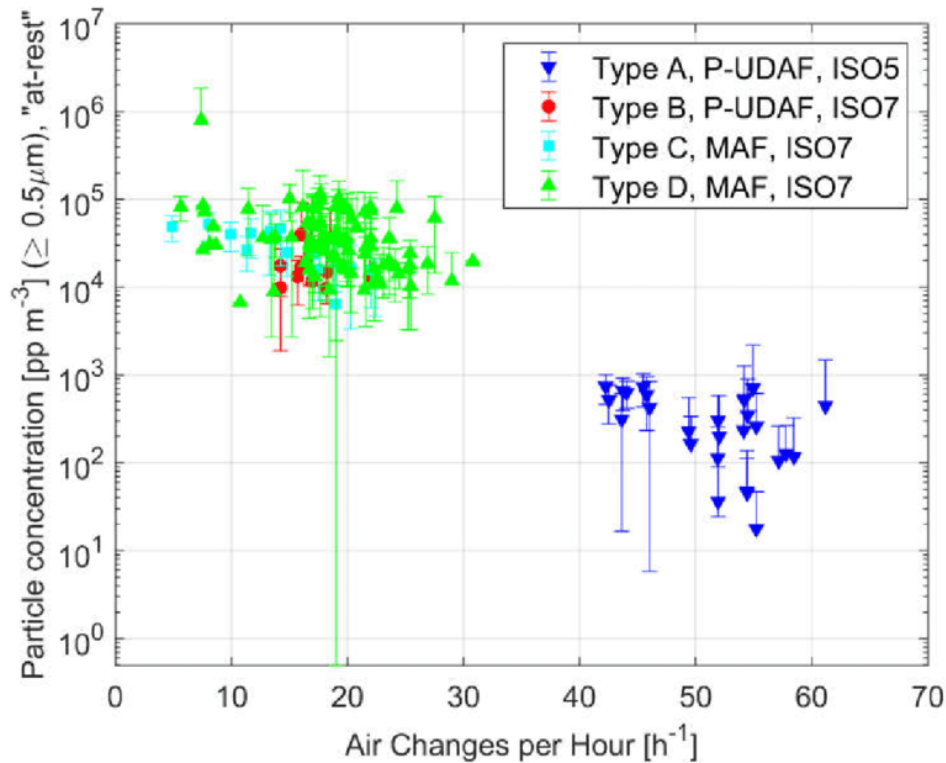
Attachment 1

“Average microbiological air contamination vs. ACH in “operational” conditions for the 175 investigated Ots” (Romano et al., 2020).



Attachment 1 continued

“Average airborne particle concentration vs. ACH in “at-rest” conditions for the 175 investigated OTs. Cumulated values for particles $\geq 0.5 \mu\text{m}$ ” (Romano et al., 2020).



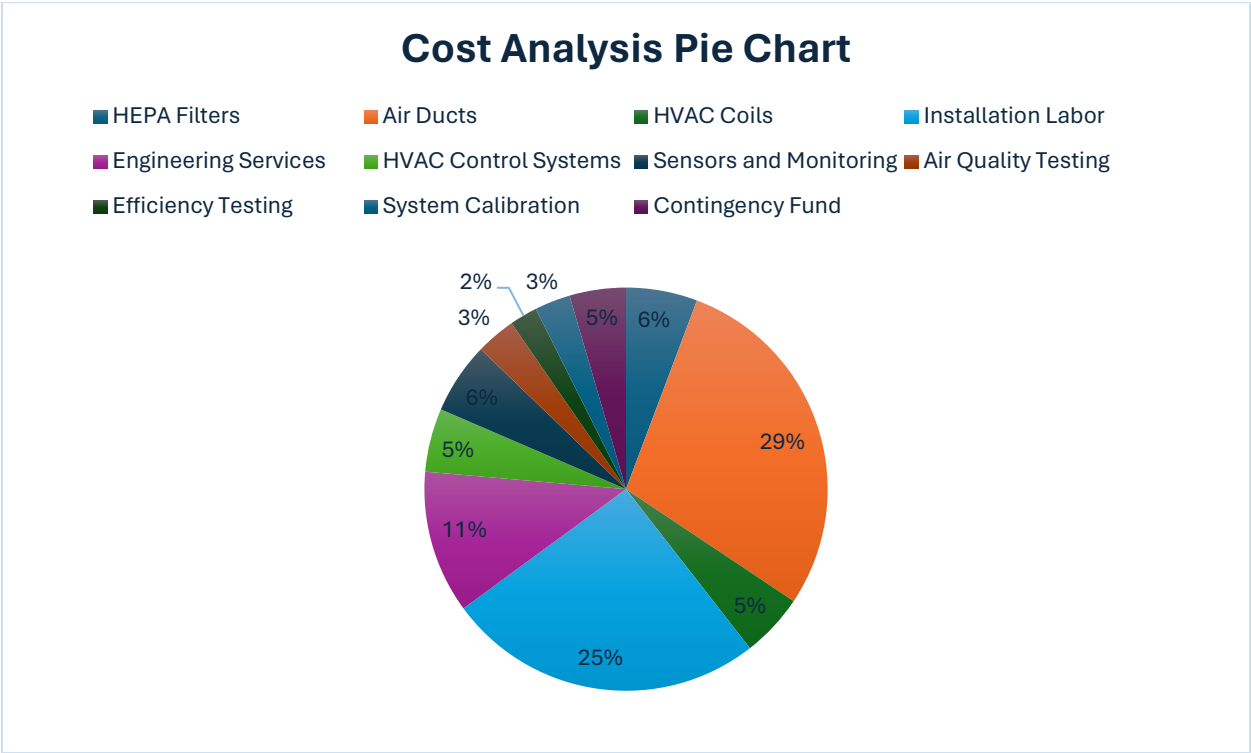
Note. Reprinted from “Operating Theatre Ventilation Systems and Their Performance in Contamination Control: “At Rest” and “In Operation” Particle and Microbial Measurements Made in an Italian Large and Multi- Year Inspection Campaign,” by F. Romano, S. Milani, R. Ricci, and C.M. Joppolo 2020, *International Journal of Environmental Research and Public Health*, 17(19), 7275.

Attachment 2

Cost Category	Description	Unit Cost	Quantity/Duration	Total Cost
Materials				
HEPA Filters	High-efficiency particulate air filters	\$600	15	\$9,000
Air Ducts	HVAC air distribution ducting	\$150/ft	300 ft	\$45,000
HVAC Coils	For heating and cooling	\$4,000	2	\$8,000
Labor				
Installation Labor	HVAC system installation	\$100/hr	400 hours	\$40,000
Engineering Services	System design and oversight	\$120/hr	150 hours	\$18,000
Equipment				
HVAC Control Systems	Automated control systems	\$8,000	1	\$8,000
Sensors and Monitoring	Environmental condition monitoring	\$300	30	\$9,000
Testing and Compliance				
Air Quality Testing	Ensuring ISO 8 standards compliance	\$500/test	10 tests	\$5,000
Efficiency Testing	System performance tests	\$700/test	5 tests	\$3,500
Commissioning				
System Calibration	Adjusting systems for optimal performance	\$1,500/day	3 days	\$4,500
Miscellaneous				
Contingency Fund	Unforeseen expenses (5%)			\$7,100
Total				\$147,100

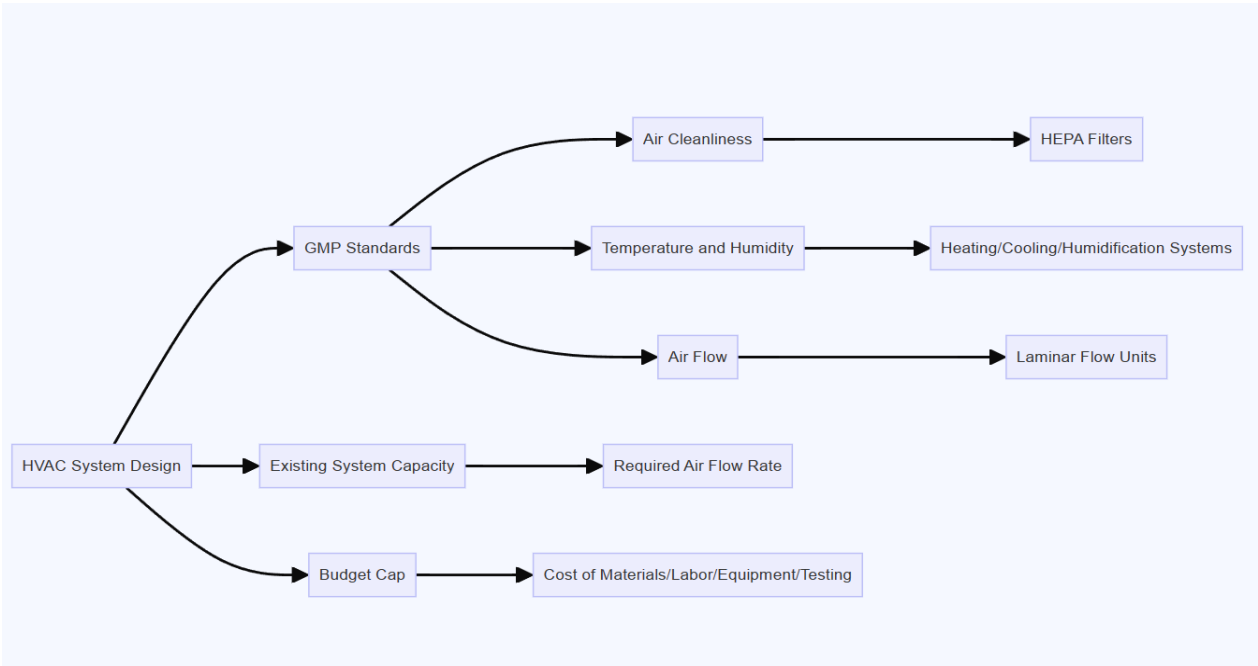
Note. This cost estimation table is a fictitious representation of the actual one, as the information cannot be disclosed due to confidentiality reasons.

Attachment 3



Note. This pie chart dataset is based on the Table.

Attachment 4



Note. This diagram of HVAC System Design Procedure is made by me.